

EE3621: Digital Signal Processing

Lecture 21: Linear-Phase FIR Filters: Characteristic Response & Z-Plane Zero Constellations (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Motivation for Linear Phase: Phase distortion in multi-carrier communication, high-fidelity audio, biomedical ECG waveforms, and radar. Definitions of Phase Delay $\tau_p(\omega)$ and Group Delay $\tau_g(\omega)$.
 - **05:00 – 14:00 (9 mins):** Mathematical Conditions for Linear Phase: Derivation of Strict Linear Phase ($\theta(\omega) = -\alpha\omega$) and Generalized Linear Phase ($\theta(\omega) = \beta - \alpha\omega$). Rigorous proof that linear phase forces impulse response symmetry ($h[n] = h[N - 1 - n]$) or antisymmetry ($h[n] = -h[N - 1 - n]$) with fixed delay $\alpha = (N - 1)/2$.
 - **14:00 – 24:00 (10 mins):** The Four Canonical Types of Linear-Phase FIR Filters: Mathematical derivations of the real amplitude function $A(\omega)$ for Types 1, 2, 3, and 4. Formal algebraic proofs for mandatory zeros at $\omega = 0$ ($z = 1$) and $\omega = \pi$ ($z = -1$), and establishing the filter selection suitability matrix.
 - **24:00 – 30:00 (6 mins):** Complex Z-Plane Zero Constellation Geometry: Reciprocal polynomial symmetry $H(z) = \pm z^{-(N-1)}H(z^{-1})$ and the fundamental theorem of reciprocal zero quadruplets $(z_0, 1/z_0, z_0^*, 1/z_0^*)$. Real axis pairs, unit-circle conjugate pairs, and boundary zero constraints.
 - **30:00 – 38:00 (8 mins):** Step-by-Step Worked Numerical Examples: Complete analytical analysis of a 5-tap Type 1 FIR filter, derivation of mandatory zeros in a Type 2 highpass impossibility proof, and reverse synthesis of an 8-tap linear-phase filter from prescribed z -plane zero roots.
 - **38:00 – 40:00 (2 mins):** Faculty Insights, Common Misconceptions, and University Semester Exam Traps.
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1 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Differentiate** between Phase Delay $\tau_p(\omega) = -\theta(\omega)/\omega$ and Group Delay $\tau_g(\omega) = -d\theta(\omega)/d\omega$, explaining how non-linear phase induces signal waveform dispersion.
2. **Derive** the necessary and sufficient time-domain conditions on the impulse response $h[n]$ that guarantee strict and generalized linear phase.

3. **Classify** any linear-phase FIR filter into Type 1, Type 2, Type 3, or Type 4 based on its length N (or order $M = N - 1$) and impulse response symmetry.
 4. **Formulate** the real-valued amplitude function $A(\omega)$ for each of the four FIR filter types.
 5. **Prove** algebraically why Type 2 filters possess a mandatory transmission zero at $\omega = \pi$, Type 3 filters at $\omega = 0$ and $\omega = \pi$, and Type 4 filters at $\omega = 0$.
 6. **Evaluate** the suitability of each filter type for Lowpass, Highpass, Bandpass, Bandstop, Differentiator, and Hilbert Transformer applications.
 7. **Map** zeros of linear-phase FIR systems in the complex z -plane, demonstrating the existence of reciprocal conjugate quadruplets.
 8. **Synthesize** linear-phase transfer functions $H(z)$ and difference equations directly from specified zero constellations.
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2. MATHEMATICAL FOUNDATIONS OF LINEAR-PHASE FIR FILTERS

2.1 Phase Distortion, Phase Delay, and Group Delay

In digital communications, audio reproduction, biomedical telemetry (such as ECG monitoring), and image processing, preserving the exact geometric shape of a waveform is often just as critical as shaping its magnitude spectrum.

When a multi-frequency signal $x[n] = \sum_k A_k \cos(\omega_k n + \phi_k)$ passes through an LTI system with frequency response $H(e^{j\omega}) = |H(e^{j\omega})|e^{j\theta(\omega)}$, each sinusoidal component experiences:

1. An amplitude scaling by $|H(e^{j\omega_k})|$.
2. A phase shift by $\theta(\omega_k)$.

To characterize the temporal delay introduced by the phase response $\theta(\omega)$, two distinct delay metrics are defined:

$$\text{Phase Delay: } \tau_p(\omega) = -\frac{\theta(\omega)}{\omega} \quad (1)$$

$$\text{Group Delay: } \tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} \quad (2)$$

- **Phase Delay** $\tau_p(\omega)$ measures the time delay experienced by individual sinusoidal carrier oscillations at frequency ω .
- **Group Delay** $\tau_g(\omega)$ measures the propagation delay experienced by the slowly varying envelope (or packet) of a group of frequencies centered around ω .

The Condition for Zero Phase Distortion: A system is said to be free of phase distortion if all spectral components undergo the **exact same time delay** α (in samples). Setting $\tau_p(\omega) = \alpha = \text{constant}$ leads directly to:

$$\theta(\omega) = -\alpha\omega \quad (3)$$

When (3) holds, the group delay is also strictly constant:

$$\tau_g(\omega) = -\frac{d}{d\omega}(-\alpha\omega) = \alpha = \tau_p(\omega) \quad (4)$$

If the phase response is non-linear, different frequency components travel at different group velocities, resulting in **dispersion** (pulse broadening, severe ringing, and mutual symbol interference), as illustrated in Figure 1.

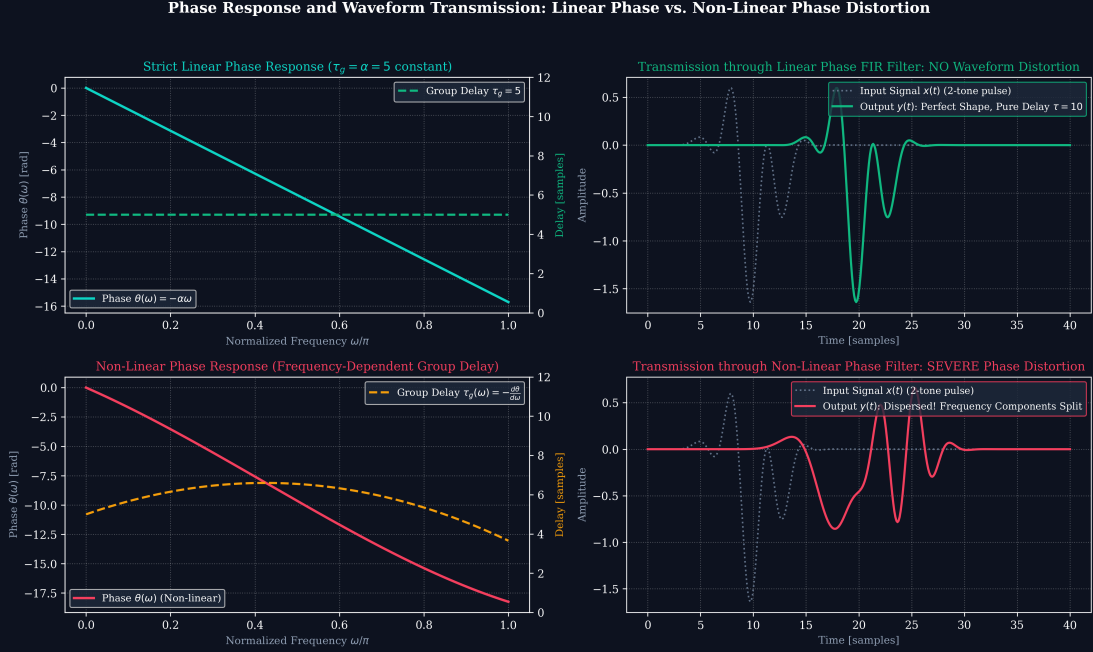


Figure 1: Phase Response and Waveform Transmission: Comparison between Strict Linear Phase (top row: constant phase and group delay $\tau = 5$, preserving wavepacket envelope shape) and Non-Linear Phase (bottom row: frequency-dependent group delay causing severe dispersion and pulse splitting).

2.2 Strict vs. Generalized Linear Phase

In digital filter theory, two classes of linear-phase systems are distinguished:

2.2.1 Class I: Strict (Pure) Linear Phase

The frequency response is expressed as:

$$H(e^{j\omega}) = A(\omega)e^{-j\alpha\omega} \quad (5)$$

where $A(\omega)$ is a strictly real-valued function of ω (which can take positive and negative values), and the phase is $\theta(\omega) = -\alpha\omega$. For this class:

$$\tau_p(\omega) = \alpha, \quad \tau_g(\omega) = \alpha \quad \forall \omega \quad (6)$$

2.2.2 Class II: Generalized Linear Phase

Here, the phase response includes a constant phase offset β :

$$\theta(\omega) = \beta - \alpha\omega \implies H(e^{j\omega}) = A(\omega)e^{j(\beta - \alpha\omega)} \quad (7)$$

In this case, the group delay remains strictly constant:

$$\tau_g(\omega) = -\frac{d\theta}{d\omega} = \alpha = \text{constant} \quad (8)$$

while the phase delay is frequency-dependent: $\tau_p(\omega) = \alpha - \beta/\omega$. Class II systems preserve the envelope shape of modulated signals (since group delay is constant) while introducing a constant 90° phase shift ($\beta = \pm\pi/2$), which is precisely required for **Hilbert transformers** and **differentiators**.

2.3 2.3 Derivation of Time-Domain Symmetry Conditions

Consider a causal FIR filter of length N (order $M = N - 1$) with real impulse response $h[n]$ for $n = 0, 1, \dots, N - 1$. Its frequency response is:

$$H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n]e^{-j\omega n} = |H(e^{j\omega})|e^{j\theta(\omega)} \quad (9)$$

We now derive the necessary condition for generalized linear phase $\theta(\omega) = \beta - \alpha\omega$.

From Euler's formula:

$$H(e^{j\omega}) = \sum_{n=0}^{N-1} h[n] \cos(\omega n) - j \sum_{n=0}^{N-1} h[n] \sin(\omega n) \quad (10)$$

Equating the phase angle:

$$\tan(\theta(\omega)) = \tan(\beta - \alpha\omega) = \frac{-\sum_{n=0}^{N-1} h[n] \sin(\omega n)}{\sum_{n=0}^{N-1} h[n] \cos(\omega n)} \quad (11)$$

Cross-multiplying and applying the trigonometric identity $\sin(A - B) = \sin A \cos B - \cos A \sin B$:

$$\begin{aligned} \sin(\beta - \alpha\omega) \sum_{n=0}^{N-1} h[n] \cos(\omega n) + \cos(\beta - \alpha\omega) \sum_{n=0}^{N-1} h[n] \sin(\omega n) &= 0 \\ \sum_{n=0}^{N-1} h[n] \sin(\beta - \alpha\omega + \omega n) &= 0 \\ \sum_{n=0}^{N-1} h[n] \sin(\beta + \omega(n - \alpha)) &= 0 \quad \forall \omega \end{aligned} \quad (12)$$

For (12) to vanish identically for all continuous frequencies $\omega \in [-\pi, \pi]$, we expand the sine term:

$$\sin \beta \sum_{n=0}^{N-1} h[n] \cos(\omega(n - \alpha)) + \cos \beta \sum_{n=0}^{N-1} h[n] \sin(\omega(n - \alpha)) = 0 \quad (13)$$

This yields two fundamental cases:

2.3.1 Case 1: $\beta = 0$ (Strict Linear Phase $\implies$ Even Symmetry)

When $\beta = 0$, $\sin \beta = 0$ and $\cos \beta = 1$, reducing the condition to:

$$\sum_{n=0}^{N-1} h[n] \sin(\omega(n - \alpha)) = 0 \quad \forall \omega \quad (14)$$

Since $\sin(\cdot)$ is an odd function, this summation cancels pairwise for all ω if and only if:

$$\alpha = \frac{N-1}{2} \quad \text{and} \quad h[n] = h[N-1-n] \quad (\text{Symmetric}) \quad (15)$$

2.3.2 Case 2: $\beta = \pm\pi/2$ (Generalized Linear Phase $\implies$ Odd Antisymmetry)

When $\beta = \pm\pi/2$, $\cos \beta = 0$ and $\sin \beta = \pm 1$, reducing the condition to:

$$\sum_{n=0}^{N-1} h[n] \cos(\omega(n - \alpha)) = 0 \quad \forall \omega \quad (16)$$

Since $\cos(\cdot)$ is an even function, this summation cancels pairwise if and only if:

$$\alpha = \frac{N-1}{2} \quad \text{and} \quad h[n] = -h[N-1-n] \quad (\text{Antisymmetric}) \quad (17)$$

Notice that for an antisymmetric impulse response with N odd, evaluating at the center index $n = \alpha = (N-1)/2$ gives:

$$h\left[\frac{N-1}{2}\right] = -h\left[N-1 - \frac{N-1}{2}\right] = -h\left[\frac{N-1}{2}\right] \implies h\left[\frac{N-1}{2}\right] = 0 \quad (18)$$

The center sample of any odd-length antisymmetric FIR filter must be identically zero!

3 THE FOUR CANONICAL TYPES OF LINEAR-PHASE FIR FILTERS

Based on the symmetry of $h[n]$ (Symmetric vs. Antisymmetric) and the parity of the filter length N (Odd vs. Even), linear-phase FIR filters are categorized into four distinct classes:

The Four Types of Linear-Phase FIR Filters: Symmetry and Amplitude Responses

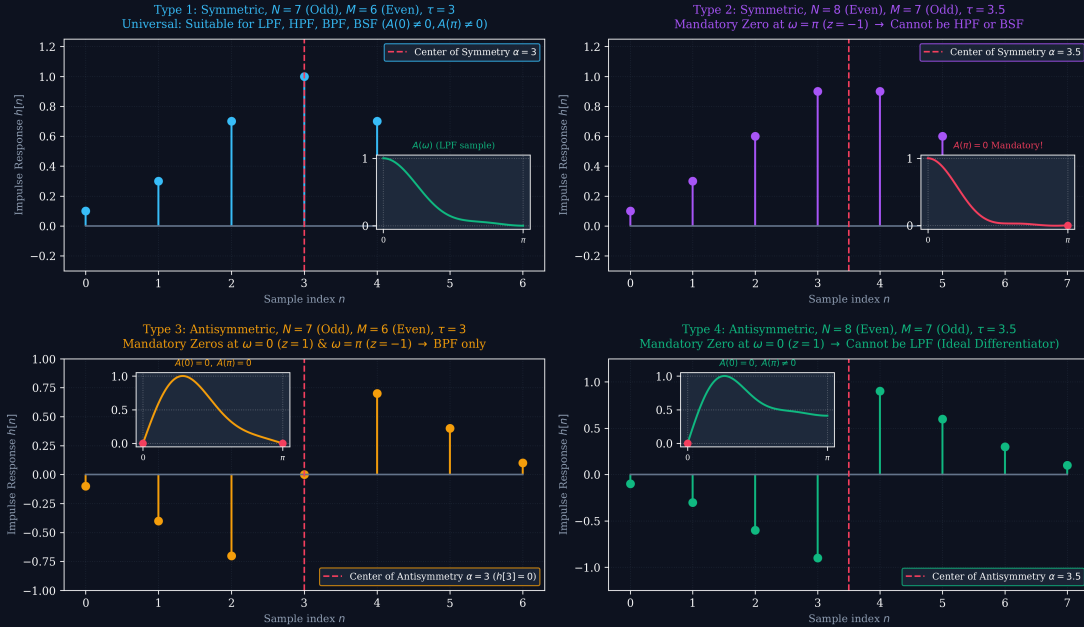


Figure 2: The Four Types of Linear-Phase FIR Filters: Time-domain impulse response stem plots showing centers of symmetry $\alpha = (N-1)/2$, accompanied by inset plots of their characteristic amplitude response $A(\omega)$.

3.1 3.1 Type 1: Symmetric Impulse Response, N Odd (M Even)

Here, $h[n] = h[N - 1 - n]$ and N is odd, meaning $\alpha = (N - 1)/2$ is an **integer**.

Frequency Response Derivation: Factoring out the pure linear-phase delay term $e^{-j\alpha\omega}$:

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=0}^{N-1} h[n] e^{-j\omega n} = e^{-j\alpha\omega} \sum_{n=0}^{N-1} h[n] e^{j\omega(\alpha-n)} \\ &= e^{-j\alpha\omega} \left[h[\alpha] + \sum_{n=0}^{\alpha-1} h[n] \left(e^{j\omega(\alpha-n)} + e^{-j\omega(\alpha-n)} \right) \right] \\ &= e^{-j\alpha\omega} \left[h[\alpha] + 2 \sum_{k=1}^{(N-1)/2} h[\alpha - k] \cos(\omega k) \right] \end{aligned} \quad (19)$$

Defining $a[0] = h[\alpha]$ and $a[k] = 2h[\alpha - k]$ for $k = 1, 2, \dots, (N - 1)/2$:

$$\mathbf{H}(e^{j\omega}) = \mathbf{e}^{-j(\frac{N-1}{2})\omega} \mathbf{A}_1(\omega), \quad \text{where} \quad \mathbf{A}_1(\omega) = \mathbf{a}[0] + \sum_{k=1}^{(N-1)/2} \mathbf{a}[k] \cos(\omega k) \quad (20)$$

Boundary Properties:

- At $\omega = 0$: $A_1(0) = a[0] + \sum a[k] = \sum_{n=0}^{N-1} h[n]$ (generally non-zero).
- At $\omega = \pi$: $A_1(\pi) = a[0] + \sum a[k](-1)^k$ (generally non-zero).

Because $A_1(0)$ and $A_1(\pi)$ are unconstrained, **Type 1 is universally suitable for Lowpass, Highpass, Bandpass, and Bandstop filters.**

3.2 3.2 Type 2: Symmetric Impulse Response, N Even (M Odd)

Here, $h[n] = h[N - 1 - n]$ and N is even, so $\alpha = (N - 1)/2$ is a **half-integer**.

Frequency Response Derivation: Grouping symmetric pairs equidistant from the half-integer center:

$$\begin{aligned} H(e^{j\omega}) &= e^{-j\alpha\omega} \sum_{n=0}^{N/2-1} h[n] \left(e^{j\omega(\alpha-n)} + e^{-j\omega(\alpha-n)} \right) \\ &= e^{-j\alpha\omega} \sum_{k=1}^{N/2} 2h \left[\frac{N}{2} - k \right] \cos \left(\omega \left(k - \frac{1}{2} \right) \right) \end{aligned} \quad (21)$$

Defining $b[k] = 2h[N/2 - k]$ for $k = 1, 2, \dots, N/2$:

$$\mathbf{H}(e^{j\omega}) = \mathbf{e}^{-j(\frac{N-1}{2})\omega} \mathbf{A}_2(\omega), \quad \text{where} \quad \mathbf{A}_2(\omega) = \sum_{k=1}^{N/2} \mathbf{b}[k] \cos \left(\omega \left(k - \frac{1}{2} \right) \right) \quad (22)$$

The Mandatory Zero Proof at $\omega = \pi$ ($z = -1$): Evaluating $A_2(\pi)$:

$$A_2(\pi) = \sum_{k=1}^{N/2} b[k] \cos \left(\pi k - \frac{\pi}{2} \right) = \sum_{k=1}^{N/2} b[k] \sin(\pi k) \equiv 0 \quad (23)$$

Alternatively, evaluating the Z-transform at $z = -1$:

$$H(-1) = \sum_{n=0}^{N-1} h[n] (-1)^{-n} = \sum_{n=0}^{N/2-1} h[n] [(-1)^{-n} + (-1)^{-(N-1-n)}] \quad (24)$$

Since $N - 1$ is odd, $(-1)^{-(N-1-n)} = (-1)^{-(N-1)}(-1)^n = (-1)(-1)^{-n} = -(-1)^{-n}$. Therefore:

$$(-1)^{-n} + (-1)^{-(N-1-n)} = (-1)^{-n} - (-1)^{-n} = 0 \implies \mathbf{H}(-1) = \mathbf{0} \quad (25)$$

Fundamental Consequence: Because $H(e^{j\pi}) = 0$ identically, a **Type 2 FIR filter can NEVER be used as a Highpass or Bandstop filter!** It is strictly restricted to Lowpass and Bandpass designs.

3.3 Type 3: Antisymmetric Impulse Response, N Odd (M Even)

Here, $h[n] = -h[N - 1 - n]$, N is odd, $\alpha = (N - 1)/2$ is an integer, and $h[\alpha] = 0$.

Frequency Response Derivation:

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=0}^{N-1} h[n] e^{-j\omega n} = e^{-j\alpha\omega} \sum_{n=0}^{\alpha-1} h[n] \left(e^{j\omega(\alpha-n)} - e^{-j\omega(\alpha-n)} \right) \\ &= e^{-j\alpha\omega} (j) \sum_{k=1}^{(N-1)/2} 2h[\alpha + k] \sin(\omega k) = e^{-j\alpha\omega} e^{j\pi/2} \sum_{k=1}^{(N-1)/2} c[k] \sin(\omega k) \end{aligned} \quad (26)$$

where $c[k] = 2h[\alpha + k]$:

$$\mathbf{H}(e^{j\omega}) = \mathbf{e}^{j(\frac{\pi}{2} - \frac{N-1}{2}\omega)} \mathbf{A}_3(\omega), \quad \text{where} \quad \mathbf{A}_3(\omega) = \sum_{k=1}^{(N-1)/2} c[k] \sin(\omega k) \quad (27)$$

Mandatory Zeros at $\omega = 0$ ($z = +1$) and $\omega = \pi$ ($z = -1$):

- At $\omega = 0$: $\sin(0) = 0 \implies \mathbf{A}_3(0) = \mathbf{0} \iff \mathbf{H}(1) = \mathbf{0}$.
- At $\omega = \pi$: $\sin(k\pi) = 0 \implies \mathbf{A}_3(\pi) = \mathbf{0} \iff \mathbf{H}(-1) = \mathbf{0}$.

Fundamental Consequence: Because it possesses mandatory zeros at both DC ($\omega = 0$) and Nyquist ($\omega = \pi$), **Type 3 CANNOT be Lowpass, Highpass, or Bandstop**. It is strictly suitable only for **Bandpass filters, Hilbert Transformers, and Differentiators**.

3.4 Type 4: Antisymmetric Impulse Response, N Even (M Odd)

Here, $h[n] = -h[N - 1 - n]$ and N is even, so $\alpha = (N - 1)/2$ is a half-integer.

Frequency Response Derivation:

$$\begin{aligned} H(e^{j\omega}) &= e^{-j\alpha\omega} (j) \sum_{k=1}^{N/2} 2h \left[\frac{N}{2} + k - 1 \right] \sin \left(\omega \left(k - \frac{1}{2} \right) \right) \\ &= e^{j(\frac{\pi}{2} - \frac{N-1}{2}\omega)} \sum_{k=1}^{N/2} d[k] \sin \left(\omega \left(k - \frac{1}{2} \right) \right) \end{aligned} \quad (28)$$

where $d[k] = 2h[N/2 + k - 1]$:

$$\mathbf{H}(e^{j\omega}) = \mathbf{e}^{j(\frac{\pi}{2} - \frac{N-1}{2}\omega)} \mathbf{A}_4(\omega), \quad \text{where} \quad \mathbf{A}_4(\omega) = \sum_{k=1}^{N/2} d[k] \sin \left(\omega \left(k - \frac{1}{2} \right) \right) \quad (29)$$

Mandatory Zero at $\omega = 0$ ($z = +1$):

$$A_4(0) = \sum_{k=1}^{N/2} d[k] \sin(0) \equiv \mathbf{0} \iff \mathbf{H}(1) = \mathbf{0} \quad (30)$$

However, at $\omega = \pi$:

$$A_4(\pi) = \sum_{k=1}^{N/2} d[k] \sin\left(\pi k - \frac{\pi}{2}\right) = \sum_{k=1}^{N/2} d[k](-1)^{k-1} \neq 0 \quad (31)$$

Fundamental Consequence: Because $H(1) = 0$, a **Type 4 FIR filter** can **NEVER** be used as a **Lowpass** or **Bandstop** filter. However, because $A_4(\pi) \neq 0$, it is the **ideal** structure for **wideband Differentiators** and **Hilbert Transformers**.

Filter Type	Symmetry	Length N	Mandatory Zeros	LPF	HPF	BPF	BSF
Type 1	Symmetric $h[n] = h[M-n]$	Odd (M Even)	None	✓	✓	✓	✓
Type 2	Symmetric $h[n] = h[M-n]$	Even (M Odd)	$z = -1$ ($\omega = \pi$)	✓	✗	✓	✗
Type 3	Antisymmetric $h[n] = -h[M-n]$	Odd (M Even)	$z = \pm 1$ ($\omega = 0, \pi$)	✗	✗	✓	✗
Type 4	Antisymmetric $h[n] = -h[M-n]$	Even (M Odd)	$z = +1$ ($\omega = 0$)	✗	✓	✓	✗

Core Rule: If $A(\pi) = 0$, filter CANNOT pass high frequencies (Forbidden for HPF, BSF).
If $A(0) = 0$, filter CANNOT pass DC/low frequencies (Forbidden for LPF, BSF). Type 1 is the ONLY universally applicable type.

Figure 3: Linear-Phase FIR Filter Suitability Matrix: Visualizing allowed (✓) and prohibited (✗) filter types based on mandatory boundary zero constraints at DC ($\omega = 0$) and Nyquist ($\omega = \pi$).

3.5 Comprehensive Summary of the Four FIR Filter Types

Table 1 provides a rigorous faculty reference for the four filter types.

Table 1: Comprehensive Architectural Audit of the Four Linear-Phase FIR Filter Types

Type	Symmetry	Length N	Delay α	Amplitude Function $A(\omega)$	Mandatory Zeros	Allowable Filter Types
Type 1	$h[n] = h[N-1-n]$	Odd (M Even)	Integer	$\sum_{k=0}^{(N-1)/2} a[k] \cos(\omega k)$	None	LPF, HPF, BPF, BSF
Type 2	$h[n] = h[N-1-n]$	Even (M Odd)	Half-Integer	$\sum_{k=1}^{N/2} b[k] \cos(\omega(k-1/2))$	$\omega = \pi$ ($z = -1$)	LPF, BPF (No HPF/BSF)
Type 3	$h[n] = -h[N-1-n]$	Odd (M Even)	Integer	$\sum_{k=1}^{(N-1)/2} c[k] \sin(\omega k)$	$\omega = 0, \pi$ ($z = \pm 1$)	BPF, Diff, Hilbert (No LPF/HPF/BSF)
Type 4	$h[n] = -h[N-1-n]$	Even (M Odd)	Half-Integer	$\sum_{k=1}^{N/2} d[k] \sin(\omega(k-1/2))$	$\omega = 0$ ($z = 1$)	HPF, BPF, Diff, Hilbert (No LPF/BSF)

4. Z-PLANE ZERO CONSTELLATION GEOMETRY

4.1 The Reciprocal Polynomial Symmetry Theorem

The Z-transform of a causal FIR filter of length N is:

$$H(z) = \sum_{n=0}^{N-1} h[n]z^{-n} \quad (32)$$

Substituting the linear-phase symmetry condition $h[n] = \pm h[N-1-n]$ (where $+$ applies to Types 1/2 and $-$ applies to Types 3/4):

$$H(z) = \sum_{n=0}^{N-1} \pm h[N-1-n]z^{-n} = \pm z^{-(N-1)} \sum_{n=0}^{N-1} h[N-1-n]z^{N-1-n} \quad (33)$$

Letting $m = N-1-n$, as n ranges from 0 to $N-1$, m ranges from $N-1$ down to 0:

$$\mathbf{H}(\mathbf{z}) = \pm \mathbf{z}^{-(N-1)} \mathbf{H}(\mathbf{z}^{-1}) \quad (34)$$

This elegant equation governs all zero locations of linear-phase FIR filters.

4.2 The Reciprocal Zero Quadruplet Theorem

Suppose z_0 is a complex zero of $H(z)$ off the real axis and not on the unit circle:

$$H(z_0) = 0, \quad \text{where } z_0 = re^{j\theta} \quad (r \neq 1, \theta \neq 0, \pi) \quad (35)$$

From fundamental complex analysis and (34), the following four zeros are guaranteed to exist simultaneously:

1. **Conjugate Zero:** Because $h[n]$ is real-valued, complex roots must occur in conjugate pairs:

$$H(z_0^*) = 0 \implies \mathbf{z}_0^* = \mathbf{r}e^{-j\theta} \quad (36)$$

2. **Reciprocal Zero:** Evaluating (34) at $z = z_0^{-1}$:

$$H(z_0^{-1}) = \pm (z_0^{-1})^{-(N-1)} H(z_0) = \pm z_0^{N-1} \cdot 0 = 0 \implies \frac{1}{\mathbf{z}_0} = \frac{1}{\mathbf{r}}e^{-j\theta} \quad (37)$$

3. **Conjugate-Reciprocal Zero:** Combining real coefficients and reciprocal symmetry:

$$\frac{1}{\mathbf{z}_0^*} = \frac{1}{\mathbf{r}}e^{j\theta} \quad (38)$$

Theorem Conclusion: Every complex zero $z_0 = re^{j\theta}$ located off the unit circle ($r \neq 1$) is accompanied by three sibling zeros, forming a **reciprocal quadruplet** in the complex z -plane:

$$\{\mathbf{z}_0, \mathbf{z}_0^*, 1/\mathbf{z}_0, 1/\mathbf{z}_0^*\} \equiv \left\{ \mathbf{r}e^{j\theta}, \mathbf{r}e^{-j\theta}, \frac{1}{\mathbf{r}}e^{-j\theta}, \frac{1}{\mathbf{r}}e^{j\theta} \right\} \quad (39)$$

3. **Real Zeros on the Unit Circle** ($z = \pm 1$): For $z = 1$, $1/z = 1$ (self-reciprocal). For $z = -1$, $1/z = -1$ (self-reciprocal). These occur as **single zeros** (or odd multiplicities):

$$z = +1 \quad (\omega = 0) \quad \text{and/or} \quad z = -1 \quad (\omega = \pi) \quad (42)$$

5. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

5.1 Example 21.1: Complete Analytical Analysis of a Type 1 FIR Filter

Problem Statement: An FIR digital filter is specified by the causal transfer function:

$$H(z) = 1 + 2z^{-1} + 3z^{-2} + 2z^{-3} + z^{-4} \quad (43)$$

1. Identify the filter type, length N , order M , and group delay τ_g .
2. Derive the analytical expression for the amplitude response $A(\omega)$.
3. Find all four zeros of $H(z)$ algebraically in the complex z -plane.
4. Verify that the zeros satisfy the reciprocal quadruplet / conjugate geometry.
5. Evaluate the filter's magnitude at $\omega = 0$, $\omega = \pi/2$, and $\omega = \pi$.

Solution:

Step 1: Classification and Group Delay The impulse response sequence is:

$$h[n] = \{1, 2, 3, 2, 1\} \quad \text{for } n = 0, 1, 2, 3, 4 \quad (44)$$

- Length $N = 5$ (Odd), Order $M = N - 1 = 4$ (Even).
- Symmetry check: $h[0] = h[4] = 1$, $h[1] = h[3] = 2$, $h[2] = 3$. Because $h[n] = h[4 - n]$, the impulse response is **strictly symmetric**.
- Therefore, this is a **Type 1 Linear-Phase FIR Filter**.
- Group delay: $\tau_g = \alpha = \frac{N-1}{2} = \frac{5-1}{2} = 2$ samples (constant for all ω).

Step 2: Amplitude Response Derivation Factoring out the delay $e^{-j2\omega}$:

$$\begin{aligned} H(e^{j\omega}) &= e^{-j2\omega} [h[2] + 2h[1] \cos(\omega) + 2h[0] \cos(2\omega)] \\ &= e^{-j2\omega} [3 + 4 \cos(\omega) + 2 \cos(2\omega)] \end{aligned} \quad (45)$$

Thus, the real-valued amplitude function is:

$$\mathbf{A}(\omega) = \mathbf{3} + \mathbf{4} \cos(\omega) + \mathbf{2} \cos(2\omega) \quad (46)$$

Step 3: Algebraic Extraction of Zeros To find the zeros, set $H(z) = 0$:

$$z^4 + 2z^3 + 3z^2 + 2z + 1 = 0 \quad (47)$$

Divide both sides by z^2 :

$$\left(z^2 + \frac{1}{z^2}\right) + 2\left(z + \frac{1}{z}\right) + 3 = 0 \quad (48)$$

Let $u = z + z^{-1}$. Then $u^2 = z^2 + 2 + z^{-2} \implies z^2 + z^{-2} = u^2 - 2$. Substituting:

$$(u^2 - 2) + 2u + 3 = 0 \implies u^2 + 2u + 1 = 0 \implies (u + 1)^2 = 0 \quad (49)$$

This gives a repeated root $u = -1$ with multiplicity 2!

Now solve for z from $z + z^{-1} = -1$:

$$z^2 + z + 1 = 0 \implies z = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2} \quad (50)$$

In polar form:

$$z_{1,2} = e^{\pm j\frac{2\pi}{3}} \quad (\text{each of multiplicity } 2) \quad (51)$$

Step 4: Verification of Zero Constellation

- Magnitude: $|z_1| = |z_2| = \sqrt{(-1/2)^2 + (\sqrt{3}/2)^2} = \sqrt{1/4 + 3/4} = 1$.
- All zeros lie **on the unit circle** at angles $\pm 2\pi/3$ ($\pm 120^\circ$).
- For unit-circle zeros, $1/z_1 = z_1^* = z_2$. The reciprocal property is self-satisfied:

$$\frac{1}{e^{j2\pi/3}} = e^{-j2\pi/3} \quad (\text{Q.E.D.}) \quad (52)$$

Step 5: Magnitude Evaluation

- At $\omega = 0$ (DC): $A(0) = 3 + 4(1) + 2(1) = \mathbf{9}$.
- At $\omega = \frac{\pi}{2}$: $A(\pi/2) = 3 + 4(0) + 2(-1) = \mathbf{1}$.
- At $\omega = \frac{2\pi}{3}$: $A(2\pi/3) = 3 + 4(-0.5) + 2(-0.5) = 3 - 2 - 1 = \mathbf{0}$ (Transmission Null).
- At $\omega = \pi$ (Nyquist): $A(\pi) = 3 + 4(-1) + 2(1) = 3 - 4 + 2 = \mathbf{1}$.

This filter exhibits lowpass-like characteristics with a double notch at $\omega = 120^\circ$.

5.2 Example 21.2: Filter Suitability Proof and Type 2 / Type 4 Analysis

Problem Statement: An electrical engineering student attempts to design a digital Highpass filter meeting the following specifications: Passband edge $\omega_p = 0.7\pi$, stopband edge $\omega_s = 0.5\pi$, using an FIR filter of length $N = 10$.

1. Can a symmetric FIR filter of length $N = 10$ fulfill this requirement? Prove your answer rigorously.
2. What is the minimum modification to N to allow a symmetric linear-phase design?
3. Could an antisymmetric filter of length $N = 10$ be used instead? Explain the engineering trade-off.

Solution:

Step 1: Analysis of Length $N = 10$ Symmetric Filter A symmetric filter with $N = 10$ has an even length ($M = 9$ odd), which is a **Type 2 FIR filter**. From (22), its amplitude response is:

$$A_2(\omega) = \sum_{k=1}^5 b[k] \cos\left(\omega\left(k - \frac{1}{2}\right)\right) \quad (53)$$

At the Nyquist frequency $\omega = \pi$:

$$A_2(\pi) = \sum_{k=1}^5 b[k] \cos\left(\pi k - \frac{\pi}{2}\right) = \sum_{k=1}^5 b[k] \sin(\pi k) \equiv 0 \quad (54)$$

Because $A_2(\pi) = 0$ identically regardless of coefficient values, the filter has an inescapable transmission zero at $\omega = \pi$. But a Highpass filter requires $|H(e^{j\pi})| \approx 1$ (unity passband gain). Therefore, **it is mathematically impossible for a Type 2 filter to function as a Highpass filter.**

Step 2: Correcting the Design via Filter Length To design a symmetric Highpass filter, the filter must be **Type 1** (N odd, M even), which has no mandatory zero at $\omega = \pi$. The student must change the length to:

$$N = 11 \quad (\text{or } N = 9) \quad (55)$$

Choosing $N = 11$ increases the order to $M = 10$ (even), creating a Type 1 filter that can freely synthesize non-zero gain at $\omega = \pi$.

Step 3: Evaluating an Antisymmetric Filter of Length $N = 10$ An antisymmetric filter of even length $N = 10$ is a **Type 4 FIR filter**.

- At $\omega = 0$: $A_4(0) = 0$ (mandatory zero, which is perfectly compatible with Highpass behavior since a Highpass filter must reject DC).
- At $\omega = \pi$: $A_4(\pi) \neq 0$ (no zero constraint, allowing passband transmission at Nyquist).

Engineering Trade-Off: While a Type 4 filter CAN achieve the magnitude response of a Highpass filter, it introduces a **constant $+90^\circ$ phase shift** ($\beta = \pi/2$), making $H(e^{j\omega}) = jA_4(\omega)e^{-j4.5\omega}$. In audio and general filtering, this $+90^\circ$ phase shift alters waveform symmetry. Hence, Type 1 with $N = 11$ is strongly preferred unless Hilbert/differential processing is specifically desired.

5.3 Example 21.3: Reverse Synthesis from Z-Plane Zero Constellation

Problem Statement: A linear-phase FIR filter is required to have:

- A complex conjugate zero pair on the unit circle at $z = e^{\pm j\pi/2} = \pm j$.
- A complex zero quadruplet off the unit circle with radius $r = 2$ and angle $\theta = \pi/3$ (60°).

Determine the filter length N , find the impulse response $h[n]$ normalized such that $h[0] = 1$, verify its linear-phase symmetry, and state the group delay.

Solution:

Step 1: Enumerating All Zeros

1. **Unit Circle Zeros:**

$$z_1 = j, \quad z_2 = -j \implies (z - j)(z + j) = z^2 + 1 \quad (56)$$

2. **Reciprocal Quadruplet Zeros ($r = 2, \theta = \pi/3$):**

$$z_3 = 2e^{j\pi/3} = 2(0.5 + j\sqrt{3}/2) = 1 + j\sqrt{3} \quad (57)$$

$$z_4 = 2e^{-j\pi/3} = 1 - j\sqrt{3} \quad (58)$$

$$z_5 = \frac{1}{2}e^{j\pi/3} = 0.5(0.5 + j\sqrt{3}/2) = 0.25 + j\frac{\sqrt{3}}{4} \quad (59)$$

$$z_6 = \frac{1}{2}e^{-j\pi/3} = 0.25 - j\frac{\sqrt{3}}{4} \quad (60)$$

Combine conjugate pairs:

$$P_{\text{outer}}(z) = (z - z_3)(z - z_4) = z^2 - 2\text{Re}(z_3)z + |z_3|^2 = z^2 - 2z + 4 \quad (61)$$

$$P_{\text{inner}}(z) = (z - z_5)(z - z_6) = z^2 - 2\text{Re}(z_5)z + |z_5|^2 = z^2 - 0.5z + 0.25 \quad (62)$$

Multiply the outer and inner quadratic factors to form the 4th-order quadruplet polynomial:

$$\begin{aligned} P_{\text{quad}}(z) &= (z^2 - 2z + 4)(z^2 - 0.5z + 0.25) \\ &= z^2(z^2 - 0.5z + 0.25) - 2z(z^2 - 0.5z + 0.25) + 4(z^2 - 0.5z + 0.25) \\ &= (z^4 - 0.5z^3 + 0.25z^2) - (2z^3 - z^2 + 0.5z) + (4z^2 - 2z + 1) \\ &= \mathbf{z^4 - 2.5z^3 + 5.25z^2 - 2.5z + 1} \end{aligned} \quad (63)$$

Notice the beautiful internal symmetry of $P_{\text{quad}}(z)$! Its coefficients are $[1, -2.5, 5.25, -2.5, 1]$.

Step 2: Multiply by the Unit Circle Quadratic Factor The overall system polynomial $P(z)$ is of 6th degree:

$$\begin{aligned} P(z) &= (z^2 + 1)(z^4 - 2.5z^3 + 5.25z^2 - 2.5z + 1) \\ &= (z^6 - 2.5z^5 + 5.25z^4 - 2.5z^3 + z^2) + (z^4 - 2.5z^3 + 5.25z^2 - 2.5z + 1) \\ &= \mathbf{z^6 - 2.5z^5 + 6.25z^4 - 5.0z^3 + 6.25z^2 - 2.5z + 1} \end{aligned} \quad (64)$$

Step 3: Express Causal System Function $H(z)$ and Impulse Response $h[n]$ Dividing by z^6 to convert to causal powers of z^{-1} :

$$H(z) = 1 - 2.5z^{-1} + 6.25z^{-2} - 5.0z^{-3} + 6.25z^{-4} - 2.5z^{-5} + z^{-6} \quad (65)$$

The impulse response sequence is:

$$\mathbf{h[n] = \{1.0, -2.5, 6.25, -5.0, 6.25, -2.5, 1.0\}} \quad (66)$$

Step 4: Symmetry and Group Delay Verification

- Length $N = 7$ (Odd), Order $M = 6$ (Even).
- Center of symmetry: $\alpha = (7 - 1)/2 = 3$.
- Check pairs: $h[0] = h[6] = 1.0$, $h[1] = h[5] = -2.5$, $h[2] = h[4] = 6.25$, center $h[3] = -5.0$.
- Because $h[n] = h[6 - n]$, this is a **Type 1 Linear-Phase FIR Filter**.
- Group delay: $\tau_g = \mathbf{3}$ samples (constant across all frequencies).

6 6. FACULTY INSIGHTS, COMMON PITFALLS & EXAM TRAPS

- **Pitfall 1: Confusing Order M with Length N in Formulas:** A filter with $N = 8$ taps has order $M = N - 1 = 7$. In university exams, students frequently write $\alpha = N/2 = 4$, which is catastrophic. The group delay is **always** $\alpha = \frac{N-1}{2}$. For $N = 8$, $\alpha = 3.5$ samples.
- **Pitfall 2: Attempting to Build a Highpass Filter with Even Length N (Type 2):** This is one of the most common exam traps. When a question asks for a Highpass filter with an even number of coefficients ($N = 10, 12, \dots$), students who attempt a standard symmetric design fail because $H(-1) = 0$ is mathematically forced. A Highpass filter with symmetric impulse response **MUST have an odd length N (Type 1)**.

- **Pitfall 3: Claiming Type 3 Filters Can Function as Lowpass Filters:** Type 3 filters have $h[n] = -h[N-1-n]$. Setting $z = 1$ ($\omega = 0$) gives $H(1) = \sum h[n] = 0$. A filter whose DC gain is zero can never be a Lowpass filter!
- **Pitfall 4: Miscounting Zeros in Reciprocal Quadruplets:** Remember that a zero on the unit circle ($|z| = 1$) generates only **2 zeros** ($e^{j\phi}, e^{-j\phi}$), NOT 4, because $1/z_0 = z_0^*$. Similarly, a real zero off the unit circle generates only **2 zeros** ($\sigma, 1/\sigma$). Only complex zeros off the unit circle generate full **4-zero quadruplets**.
- **Pitfall 5: Non-Causal vs Causal Symmetry Centers:** For a non-causal zero-phase filter, $h[-n] = h[n]$ and $\alpha = 0$. For a causal physical filter, $h[n] = h[N-1-n]$ and the center of symmetry is delayed to $\alpha = (N-1)/2$.

7. QUICK REVISION & SUMMARY CHEAT-SHEET

Table 2: Linear-Phase FIR Filters Quick Reference Reference Sheet

Concept / Parameter	Mathematical Definition / Operational Formula
Phase Delay	$\tau_p(\omega) = -\theta(\omega)/\omega$
Group Delay	$\tau_g(\omega) = -d\theta(\omega)/d\omega$
Strict Linear Phase	$\theta(\omega) = -\alpha\omega \implies h[n] = h[N-1-n], \quad \alpha = (N-1)/2$
Generalized Linear Phase	$\theta(\omega) = \beta - \alpha\omega \implies h[n] = \pm h[N-1-n], \quad \beta = \pm\pi/2$
Type 1 FIR	Symmetric, N Odd (M Even) $\implies$ Universal (LPF, HPF, BPF, BSF)
Type 2 FIR	Symmetric, N Even (M Odd) $\implies$ Inherent zero at $\omega = \pi$ ($z = -1$) (No HPF/BSF)
Type 3 FIR	Antisymmetric, N Odd (M Even) $\implies$ Inherent zeros at $\omega = 0, \pi$ ($z = \pm 1$) (BPF only)
Type 4 FIR	Antisymmetric, N Even (M Odd) $\implies$ Inherent zero at $\omega = 0$ ($z = 1$) (Ideal Differentiator)
Polynomial Symmetry	$H(z) = \pm z^{-(N-1)} H(z^{-1})$
Reciprocal Quadruplet	If $z_0 = re^{j\theta}$ is a zero, $\{re^{j\theta}, re^{-j\theta}, \frac{1}{r}e^{-j\theta}, \frac{1}{r}e^{j\theta}\}$ are all zeros
Unit Circle Zeros	Self-reciprocal pairs $e^{\pm j\phi}$ (provide stopband transmission nulls)